

Homework 6, Question 4c Explanation

Poisson Process with Random Classification

1 Problem Statement

Given:

- $\{N_t\}$ is a Poisson process with rate λ
- Each arrival is **independently** classified as:
 - Woman with probability $\frac{1}{2}$
 - Man with probability $\frac{1}{2}$
- The genders are independent of each other and of the arrival times
- M_t = number of women that arrived by time t

Find: $\mathbb{E}(N_t \mid M_t = m)$

2 Key Concept: Poisson Thinning

Poisson Thinning Theorem

When you “thin” a Poisson process with rate λ by randomly and independently classifying each arrival with probability p :

- The “successes” form a Poisson process with rate $p\lambda$
- The “failures” form a Poisson process with rate $(1 - p)\lambda$
- **These two processes are independent**

2.1 Applying Thinning to Our Problem

In this problem, we’re classifying each arrival as woman (probability $\frac{1}{2}$) or man (probability $\frac{1}{2}$):

$$M_t = \text{number of women by time } t \sim \text{Poisson}\left(\frac{\lambda t}{2}\right)$$

$$W_t = \text{number of men by time } t \sim \text{Poisson}\left(\frac{\lambda t}{2}\right)$$

$$N_t = M_t + W_t \quad (\text{total arrivals})$$

Crucial Fact: M_t and W_t are **independent** random variables.

3 Solution

We want to find $\mathbb{E}(N_t \mid M_t = m)$.

Since $N_t = M_t + W_t$, we can write:

$$\begin{aligned}\mathbb{E}(N_t \mid M_t = m) &= \mathbb{E}(M_t + W_t \mid M_t = m) \\ &= \mathbb{E}(M_t \mid M_t = m) + \mathbb{E}(W_t \mid M_t = m)\end{aligned}\tag{1}$$

3.1 Step 1: Evaluate $\mathbb{E}(M_t \mid M_t = m)$

This is straightforward:

$$\mathbb{E}(M_t \mid M_t = m) = m$$

If we know $M_t = m$, then the expected value of M_t is simply m .

3.2 Step 2: Evaluate $\mathbb{E}(W_t \mid M_t = m)$

Key Step: Using Independence

Since M_t and W_t are **independent**, conditioning on $M_t = m$ does not affect the distribution of W_t :

$$\mathbb{E}(W_t \mid M_t = m) = \mathbb{E}(W_t)$$

Therefore:

$$\mathbb{E}(W_t) = \frac{\lambda t}{2}$$

This is just the mean of a Poisson random variable with parameter $\frac{\lambda t}{2}$.

3.3 Step 3: Combine the Results

Substituting back into equation (1):

$$\begin{aligned}\mathbb{E}(N_t \mid M_t = m) &= m + \mathbb{E}(W_t) \\ &= m + \frac{\lambda t}{2}\end{aligned}$$

Final Answer

$$\boxed{\mathbb{E}(N_t \mid M_t = m) = m + \frac{\lambda t}{2}}$$

Alternatively, since $\mathbb{E}(M_t) = \frac{\lambda t}{2}$, we can write:

$$\boxed{\mathbb{E}(N_t \mid M_t = m) = m + \mathbb{E}(M_t)}$$

4 Intuition

Let's interpret what this result means:

“Given that we observed m women, how many total people do we expect?”

Answer:

- We know **for certain** there were m women
- The number of men is **independent** of the number of women (because each person's gender is determined independently)
- So we expect the **average** number of men: $\mathbb{E}(W_t) = \frac{\lambda t}{2}$
- Total expected = $m + \frac{\lambda t}{2}$

4.1 Why Independence Matters

The key insight is that **independence** means:

Knowing $M_t = m$ gives us no information about W_t

Therefore:

$$\mathbb{E}(W_t \mid M_t = m) = \mathbb{E}(W_t)$$

If M_t and W_t were *not* independent, we couldn't make this simplification!

5 Numerical Example

Example: Suppose $\lambda = 4$ arrivals per hour, $t = 2$ hours, and we observe $m = 3$ women.

What is the expected total number of arrivals?

Solution:

$$\begin{aligned}\mathbb{E}(N_t \mid M_t = 3) &= 3 + \frac{\lambda t}{2} \\ &= 3 + \frac{4 \times 2}{2} \\ &= 3 + 4 \\ &= 7 \text{ arrivals}\end{aligned}$$

Interpretation:

- We know there were 3 women
- On average, we expect $\frac{4 \times 2}{2} = 4$ men
- Total expected: $3 + 4 = 7$ people

6 Common Mistakes to Avoid

6.1 Mistake 1: Forgetting Independence

Wrong: “Since we saw more women than expected, there must be fewer men”

Correct: The number of men is independent of the number of women, so observing $M_t = m$ doesn't change our expectation for W_t .

6.2 Mistake 2: Using $\mathbb{E}(N_t)$ Instead

Wrong: $\mathbb{E}(N_t \mid M_t = m) = \mathbb{E}(N_t) = \lambda t$

Correct: Conditioning on $M_t = m$ changes the expectation. We need to account for the fact that we *know* there are m women.

6.3 Mistake 3: Double Counting

Wrong: $\mathbb{E}(N_t \mid M_t = m) = m + \lambda t$ (counting m women plus all expected arrivals)

Correct: We add m (known women) plus $\frac{\lambda t}{2}$ (expected men only), not all arrivals.

7 Related Facts

7.1 Conditional Distribution

Not only is the conditional *expectation* straightforward, but the full conditional *distribution* is also simple:

$$(N_t \mid M_t = m) \sim m + \text{Poisson}\left(\frac{\lambda t}{2}\right)$$

In other words, N_t given $M_t = m$ is distributed as m (deterministic) plus an independent Poisson random variable with rate $\frac{\lambda t}{2}$.

7.2 Variance

What about $\text{Var}(N_t \mid M_t = m)$?

$$\begin{aligned}\text{Var}(N_t \mid M_t = m) &= \text{Var}(M_t + W_t \mid M_t = m) \\ &= \text{Var}(m + W_t \mid M_t = m) \\ &= \text{Var}(W_t \mid M_t = m) \\ &= \text{Var}(W_t) \quad (\text{by independence}) \\ &= \frac{\lambda t}{2}\end{aligned}$$

So the conditional variance equals the variance of W_t , which makes sense since m is constant when conditioning on $M_t = m$.

8 Summary

Key Takeaways

1. **Poisson thinning** splits a Poisson process into independent components
2. When random variables are **independent**, conditioning on one doesn't affect the other
3. The answer $m + \frac{\lambda t}{2} = (\text{known women}) + (\text{expected men})$
4. This generalizes: if probability of woman is p , then

$$\mathbb{E}(N_t \mid M_t = m) = m + (1 - p)\lambda t$$